

Designing Bioinspired Anti-Biofouling Surfaces based on a Superwettability Strategy

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Anti-biofouling surfaces are of high importance owing to their crucial roles in biosensors, biomedical devices, food processing, the marine industry, etc. However, traditional anti-biofouling surfaces based on either the release of biocidal compounds or surface chemical/physical design cannot satisfy the practical demands when meeting real-world complex conditions. The outstanding performances of natural anti-biofouling surfaces motivate the development of new bioinspired anti-biofouling surfaces. Herein, a novel strategy is proposed for rationally designing bioinspired anti-biofouling surfaces based on superwettability. By utilizing the trapped air cushions or liquid layers, Lotus leaf inspired superhydrophobic surfaces, fish scales inspired underwater superoleophobic surfaces, and Nepenthes pitcher plants inspired omniphobic slippery surfaces have been successfully designed as anti-biofouling surfaces to effectively resist proteins, bacteria, cells, and marine organisms. It is believed that these novel superwettability-based anti-biofouling surfaces will bring a new era to both biomedical technology and the marine industry, and will greatly benefit human health and daily life in the near future.

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1. Introduction

Biofouling, the attachment and accumulation of undesired biomolecules, bacteria, cells, organisms, and their deposits on material surfaces, is an ubiquitous problem in a variety of practical applications, such as biosensors, biomedical devices, food processing, and marine industry.^[1–3] Most of the current anti-biofouling strategies rely on the release of biocidal compounds or surface chemical/physical design. For example, by designing coatings that release biocidal compounds^[4,5] such as antibiotics,^[6] quaternary ammonium salts,^[7] or silver ions^[8] into the surrounding environments, fouling organisms can be effectively killed. Another strategy is based on the utilization of either hydrophilic polymers or zwitterionic molecules to prevent the adhesion of fouling organisms.^[9] Owing to the steric exclusion effect and surface hydration layer, hydrophilic polymeric materials, such as poly(ethylene glycol) (PEG),^[10,11] poly(ethylene oxide) (PEO),^[12] and poly(2-hydroxyethyl methacrylate) (HEMA),^[13] have shown great capability to resist nonspecific adsorption of proteins even from undiluted blood plasma and reduce bacterial attachment and biofilm formation. Zwitterionic surfaces,^[14–16]

including carboxybetaine, sulfobetaine, and phosphorylcholine, have emerged as a robust ultra-low-fouling platform. The key to their outstanding anti-biofouling properties is the nanometer-scale homogenous mixture of balanced charge groups since both are strongly hydrated through ionic solvation. In addition, physical strategy, in particular engineering surface topographies, such as channels, pillars, ribs and hierarchically wrinkles with appropriate topographical features, provides an alternative method for anti-biofouling by affecting the topographical interactions between fouling organisms and material surfaces.^[17–20] However, these strategies have substantial weaknesses. For instance, owing to the sustained release of biocidal compounds, this strategy usually causes serious environmental and health problems. The hydrophilic polymeric surfaces suffer from the failure of the chemical groups (e.g., oxidation of hydroxyl group of PEG coatings), limiting their long-term performance. The zwitterionic surfaces may lose their efficacy in complex environments, such as seawater with high salinity. The textured surfaces can only inhibit the attachment of limited fouling species due to the diverse shapes and sizes of fouling organisms. Therefore, it is still challenging to develop novel anti-biofouling surfaces with universal, effective, environmentally benign, low-cost, and long-term durable characters to satisfy the practical demands.

Although the fouling processes are very complex and may vary from one species to another, fouling of a new surface is typically described as following process: adsorption of proteins or polysaccharides, formation of a conditioning layer, primary colonization by organisms such as bacteria and diatoms, and finally formation of bacterial biofilms or mature biofouling community.^[2,3,21,22] During the fouling processes, one significant step is the contact and formation of initial bonds between fouling organisms and material surfaces, which is actually mediated by the contact between aqueous solutions and material surfaces since fouling organisms usually occur in aqueous solutions.^[23] Thus, the contact step is closely associated with the wettability of material surfaces, indicating that regulating the surface wettability might be a promising way to control biofouling. Numerous superwettability-based platforms have been adopted by natural creatures to achieve outstanding anti-biofouling performances. One of the most famous examples is the self-cleaning Lotus leaf with superhydrophobic property, which is attributed to the cooperation between hierarchical micro/nanostructures and low surface energy waxes (**Figure 1a**).^[24–26] The superhydrophobic surfaces, which can trap a stable air cushion in the spaces of the micro/nanostructures, result in the significant reduction of contact between water droplets and material surfaces, thus preventing the adhesion of fouling organisms and washing out the contaminations with water droplets rolling on the surfaces.^[24] Another example is the self-cleaning fish scales with underwater superoleophobic property (**Figure 1b**).^[27] By trapping a water layer on the surfaces, the underwater superoleophobic surfaces can minimize the contact between oil droplets and material surfaces, thereby significantly inhibiting the adhesion of oil pollution and fouling organisms. Employing the same strategy as the underwater fish scales, the *Nepenthes* pitcher plants inspired slippery liquid-infused



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porous surfaces (SLIPS) have been also developed and shown powerful capability to repel various liquids in air (**Figure 1c**).^[28–30] By infiltrating low-surface-energy porous solids with lubricating liquids and forming a stable, immobilized, and smooth liquid-like omniphobic surface, immiscible liquids deposited on the SLIPS can be easily removed even under weak shear forces, thus provide great promising for the resistance of fouling organisms. In this concept paper, we will introduce the superwettability-based strategy for the rational design of bioinspired anti-biofouling surfaces. We will focus on the recent development of superhydrophobic, underwater superoleophobic, and omniphobic slippery surfaces towards universal, effective, environmentally benign, and long-term anti-biofouling.

2. Bioinspired Superhydrophobic Surfaces

Bioinspired superhydrophobic surfaces have been developed as effective platform for anti-biofouling via dramatically reducing the contact between fouling organisms and material surfaces. To verify the crucial role of superhydrophobicity in anti-biofouling, Ishizaki et al. studied the adhesion behaviors of NIH 3T3 fibroblast cells on superhydrophobic and superhydrophilic surfaces.^[32] As shown in **Figure 2a**, compared with the superhydrophilic regions, the superhydrophobic regions can greatly reduce the adhesion, growth, and proliferation of cells after cultured for 1 h and 24 h. Recently, Dou et al. demonstrated that the adhesion of bacteria can be regulated by tuning the wettability of hierarchically structured surfaces.^[33]

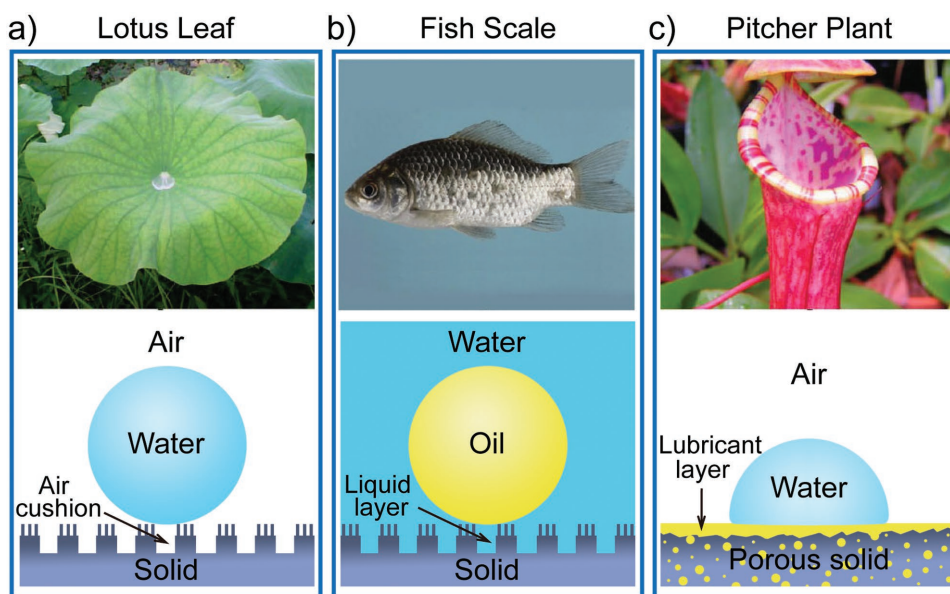


Figure 1. Natural anti-biofouling surfaces with superwettability and their design principles. a) Lotus leaf with superhydrophobicity and low adhesion in air. b) Fish scales with superoleophobicity and low adhesion under water. c) Pitcher plant with omniphobicity in air. By using the trapped air cushions or liquid layers, these surfaces can significantly prevent the closely contact and adhesion between liquid droplets and material surfaces, thus leading to the excellent anti-biofouling performances. Reproduced with permission.^[31] Copyright 2016, American Chemical Society.

By regulating the ratio of hydrophilic and hydrophobic precursors, they fabricated the hierarchically structured surfaces with tunable wettability, including superhydrophilicity, moderate wettability, and superhydrophobicity. By investigating the dynamic adsorption behavior of peptidoglycan (PGN), whose adsorption was mediated by surface wettability, the adhesion mechanism of Gram-positive bacteria *Micromonospora purpurea* (*M. purpurea*) was explored on these surfaces. They revealed that the moderate wettable surfaces showed the highest adhesion level of *M. purpurea* due to the strong hydrogen bonds or hydrophobic interaction between PGN and the surfaces. Significant decrease of *M. purpurea* adhesion was observed on the superhydrophilic (83.4%) and superhydrophobic (88.6%) surfaces because the bound water or trapped air can greatly minimize the hydrogen bonds or hydrophobic interaction between PGN and surfaces.

The adhesion behaviors of fouling organisms were also investigated on switchable superhydrophobic surfaces.^[37,34] For example, through combining colloidal lithography with electrodeposited polythiophene derivatives, Pernites et al. developed a colloiddally templated superhydrophobic polythiophene surfaces with potential-induced reversible wettability between hydrophilicity and superhydrophobicity (Figure 2b).^[34] The doped hydrophilic polythiophene film leads to the increased attachment of both protein and *Escherichia coli* (*E. coli*). Applying a low potential (1.05 V), the undoped and superhydrophobic polythiophene film can efficiently inhibit the adhesion of fibrinogen proteins and *E. coli*, indicating the superior anti-biofouling capability of superhydrophobic surfaces.

Bioinspired superhydrophobic surfaces have been also revealed as an efficient way to resist the adhesion of platelets.^[35,38] The adhesion and activation of platelets on material surfaces may result in blood coagulation and thromboses,

which affect the successful application of artificial implantations and other blood-contacting medical devices. By dip-coating fluorinated poly(carbonate urethane) onto aligned carbon nanotube films, Sun et al. reported blood-compatible superhydrophobic surfaces for in vitro adhesion test of platelets (Figure 2c).^[35] Compared with the smooth films, the superhydrophobic films can largely decrease the adhesion of platelets and avoid the activation and spreading of the occasionally attached platelets. In addition, hierarchical superhydrophobic surfaces, fabricated by a layer-by-layer particle deposition approach, can significantly reduce the adsorption of proteins (up to 90% decrease) and completely suppress the adhesion and activation of platelets.^[38]

Various bioinspired superhydrophobic surfaces have been developed for the anti-adhesion of bacteria and cells.^[39–46] For example, Privett et al. reported a silica-colloid-based superhydrophobic coating by using well-defined sol-gel chemistry. Such superhydrophobic coating can reduce the adhesion of highly pathogenic *Staphylococcus aureus* (*S. aureus*) and *Pseudomonas aeruginosa* (*P. aeruginosa*) (ca. 2 orders of magnitude versus controls), making them excellent candidates for medical devices.^[42] The highly inhibition of gram-negative *E. coli* can be also effectively realized by using shrink-induced superhydrophobic consumer plastics (PS, PC, and PE)^[43] and superamphiphobic cellulose nanofiber surfaces.^[45] In addition, through one-step electrospraying poly(L-lactide) (PLLA) and modifying silica nanoparticles, Hu et al. fabricated biodegradable superhydrophobic coatings, which significantly reduced the adhesion of C3H10T $\frac{1}{2}$ cell and *S. aureus* compared with the electrosprayed PLLA coating and PLLA cast films.^[46] Owing to their biodegradability, the superhydrophobic coatings may potentially be a platform for preparing a temporary anti-biofouling coating for surgical instruments. More

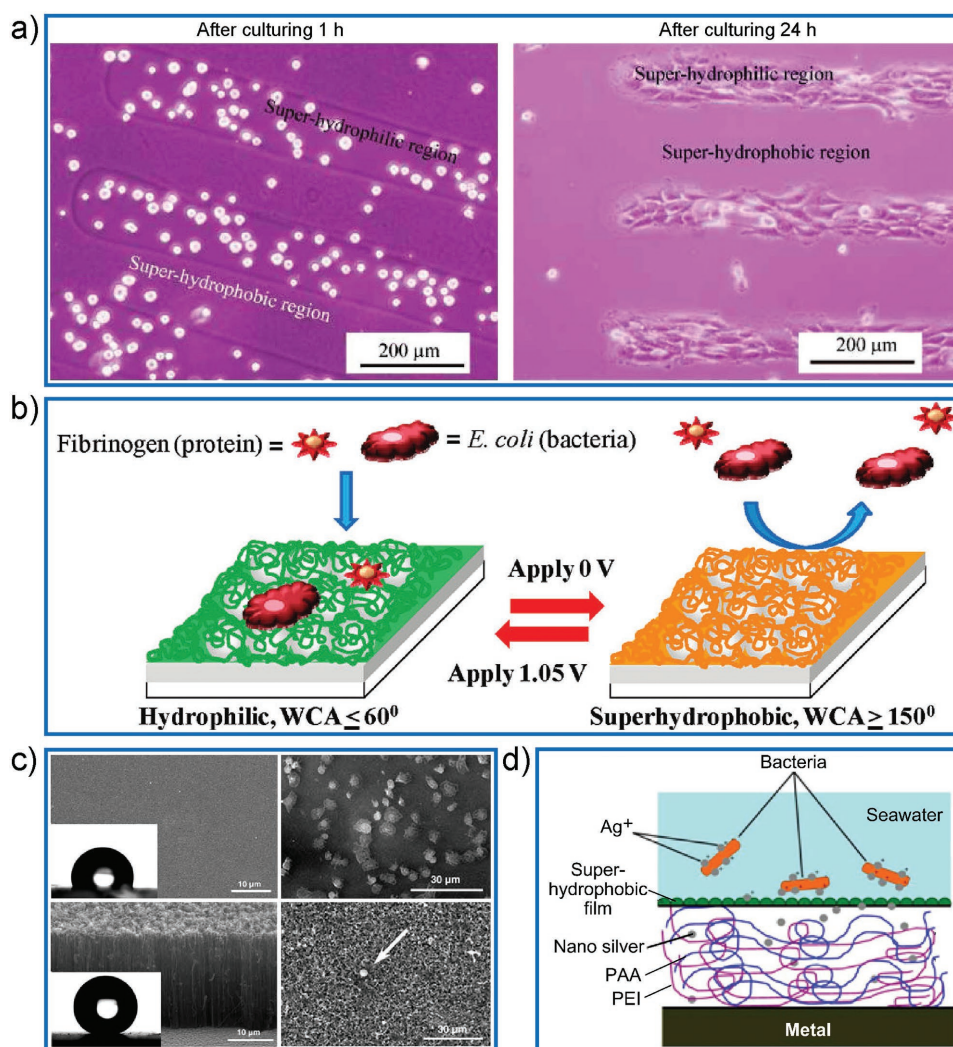


Figure 2. Bioinspired superhydrophobic surfaces for anti-biofouling. a) Compared with superhydrophilic regions, superhydrophobic regions can greatly reduce the adhesion, growth, and proliferation of cells. Reproduced with permission.^[32] Copyright 2010, American Chemical Society. b) Electro-responsive superhydrophobic surfaces can be used to control the adhesion of fibrinogen proteins and *E. coli*. Reproduced with permission.^[34] Copyright 2012, American Chemical Society. c) Compared with the smooth films, the nanostructured superhydrophobic films can largely decrease the adhesion of platelets and avoid the activation and spreading of the occasionally attached platelets. Reproduced with permission.^[35] Copyright 2005, John Wiley & Sons, Inc. d) Superhydrophobic surfaces combined with Ag nanoparticles can inhibit fouling organisms and kill the occasionally attached fouling organisms simultaneously. Reproduced with permission.^[36] Copyright 2012, American Chemical Society.

recently, Pechook et al. reported novel passive superhydrophobic surfaces fabricated via the self-assembly of paraffin or fluorinated wax crystals for long-term preventing the attachment of biofilms.^[47] They demonstrated that the wax modified surfaces passively (with no toxic influence on bacterial cells) inhibited the formation of biofilms (reduction of 95.6–99.9% over a 7 day period) by *B. cereus* and *P. aeruginosa*. Importantly, the wax surfaces can be formed on a variety of materials and intricately shaped surfaces, showing potentially feasible for various medical and industrial applications. However, Fadeeva et al. revealed that the bacterial adhesion on superhydrophobic titanium surfaces is highly dependent on the nature of the bacteria.^[41] They observed that the rod-shaped *P. aeruginosa* was not able to colonize these superhydrophobic titanium surfaces. However, the spherical *S. aureus* could colonize the same surfaces,

suggesting that the shape of the bacteria might affect their adhesion.

Superhydrophobic surfaces combined with biocides have been also developed to inhibit biofouling and kill the occasionally attached biofouling simultaneously (Figure 2d).^[36,48,49] A superhydrophobic coating immobilized with silver nanoparticles was fabricated to prevent the adhesion of bacteria and marine fouling.^[36] Due to the presence of air cushions between the substrate and seawater, the superhydrophobic surfaces can slow down the release of Ag ions, thus significantly prolonging the duration of sustained release of antibacterial Ag ions.^[49] Compared to the traditional anti-biofouling coatings composed of silver nanoparticles, these superhydrophobic coatings can significantly improve the antibacterial performance and long-term effectiveness.

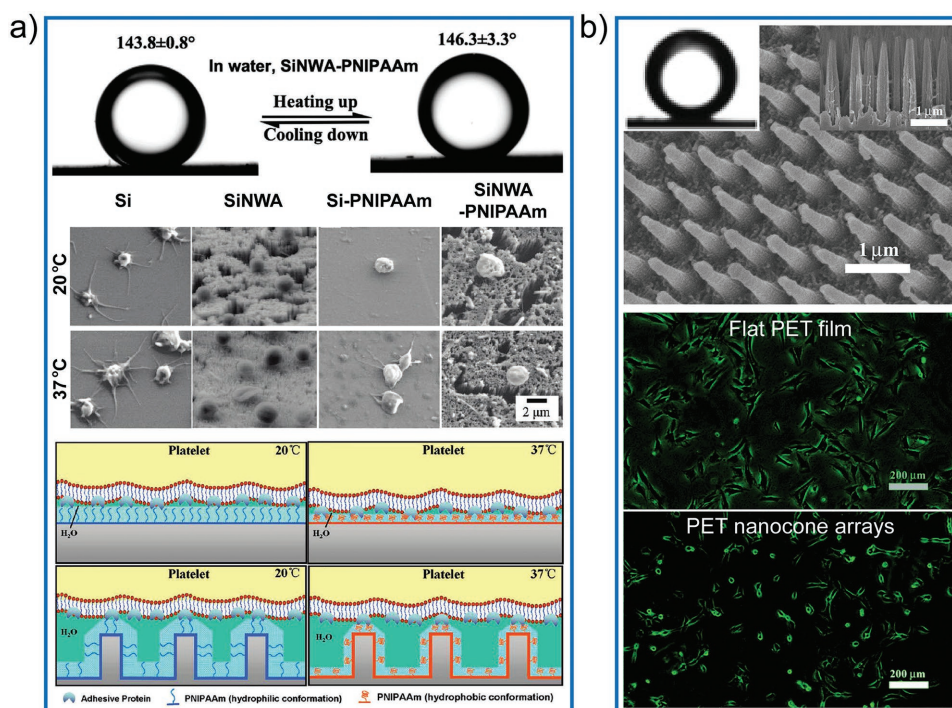


Figure 3. Bioinspired underwater superoleophobic surfaces for anti-biofouling. a) The underwater superoleophobic SiNW-PNIPAAm surfaces exhibit excellent anti-platelet adhesion in vitro both below and above the lower critical solution temperature (LCST) of PNIPAAm ($\approx 32^\circ\text{C}$). Reproduced with permission.^[57] Copyright 2009, American Chemical Society. b) Underwater superoleophobic PET nanocone arrays can weaken the interactions between the surfaces and secreted cell-adhesion proteins, thus decreasing the cells anchored to the surfaces. Reproduced with permission.^[58] Copyright 2014, Royal Society of Chemistry.

3. Bioinspired Underwater Superoleophobic Surfaces

Boats are contaminated by plankton in seawater. In oil-polluted water, seabirds are endangered by oil pollution, while fishes can keep their body clean under water. In fact, marine creatures, such as carps,^[27] pilot whales,^[50] sharks,^[51] filefishes,^[52] and seaweed,^[53] exhibit reliable strategies against unwanted growth of marine organisms. For example, the scales of carps are composed of hydrophilic composition (including calcium phosphate, protein, and a thin layer of mucus) and oriented micropapillae with nanostructures. Such scales can trap water layer on their surfaces, resulting in the underwater superoleophobicity, thereby dramatically reducing the contact between biofouling and material surfaces.^[27] Despite spending their entire lives submerged, sharks can resist largely adhesion of plants and barnacles, attributing to their underwater anisotropic superoleophobicity induced by the secreted mucus and riblet structures.^[51] In addition, as a kind of marine plants, seaweeds, which possess rich content of polysaccharides, show salt-tolerant underwater superoleophobicity, leading to their remarkable anti-biofouling capability.^[53]

Inspired by these interesting phenomena, a novel strategy based on underwater superoleophobicity has been developed and exhibited great potential for efficient anti-biofouling.^[27,54–56] For example, by grafting poly(N-isopropylacrylamide) (PNIPAAm) on silicon nanowire arrays (SiNW) via surface-initiated atom transfer radical polymerization

(SI-ATRP), Chen et al. reported a thermal-responsive nanostructured (SiNW-PNIPAAm) surface, showing underwater superoleophobic property (**Figure 3a**).^[57] The SiNW-PNIPAAm surfaces exhibit excellent anti-adhesion of platelets in vitro both below and above the lower critical solution temperature (LCST: critical temperature below which a mixture is miscible) of PNIPAAm ($\approx 32^\circ\text{C}$), while the smooth PNIPAAm surfaces can largely reduce the adhesion of platelets only below the LCST. This is distinct from the common view on the PNIPAAm film, which is hydrophobic above the LCST and hydrophilic below the LCST. The SiNW-PNIPAAm surfaces exhibit underwater superoleophobicity and low adhesion to oil droplets at both 20°C and 37°C , indicating that the nanostructured surfaces can keep a relatively high ratio of water content, which is responsible for the largely reduced adhesion of platelets. These results provide a new underwater superoleophobic strategy for developing blood-compatible and anti-biofouling surfaces.

The anti-cell adhesion of underwater superoleophobic surfaces has been also reported. Liu et al. presented that bioinspired polyethylene terephthalate (PET) nanocone arrays exhibited underwater superoleophobicity with oil contact angle of 171.8° .^[58] After culturing for 3 days, most of the mouse MC3T3-E1 osteoblast cells displayed a circle-like or short rod-like shape but not the highly spread morphology on the PET nanocone arrays, indicating that the cell adhesion could be greatly inhibited (**Figure 3b**). In contrast, the cells adhered well and performed great degree of spreading onto the flat PET film, indicating that the cells maintained

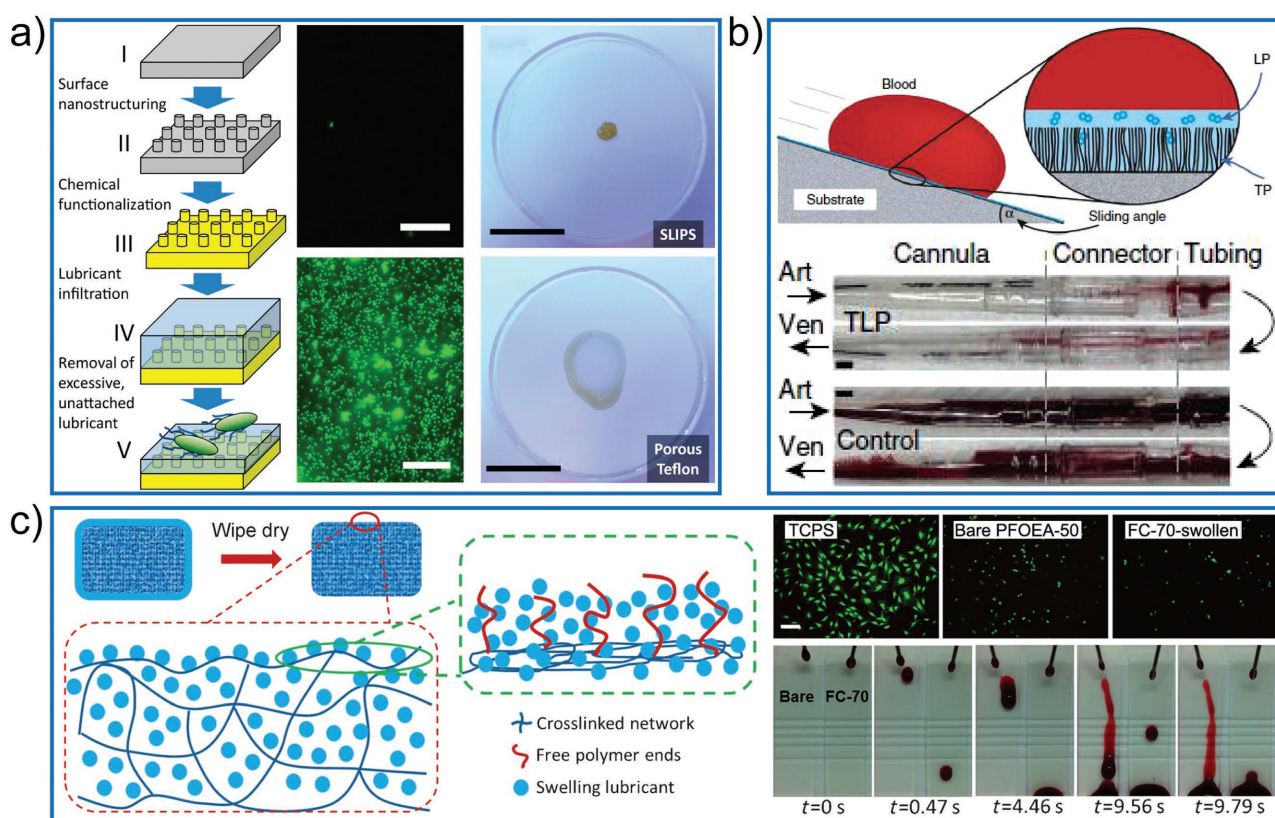


Figure 4. Bioinspired slippery surfaces for anti-biofouling. a) Compared with the porous PTFE that accumulate biofilm within hours, lubricant infused porous polytetrafluoroethylene (PTFE) based slippery surfaces can suppress 96%–99% of bacteria over at least a 7 days period under both static and physiologically realistic flow conditions. Reproduced with permission.^[60] Copyright 2012, National Academy of Sciences. b) The tethered-liquid perfluorocarbon (TLP) based slippery surfaces exhibit a remarkable decrease in occlusive thrombosis in both the arterial and venous sections when directly contact with whole blood flowing at high, physiologically relevant rates for at least 8 h in vivo. Reproduced with permission.^[61] Copyright 2014, Nature Publishing Group. c) Lubricant-swollen fluorogels based slippery surfaces can significantly inhibit the cell spreading and minimize the adhesion of blood. Reproduced with permission.^[62] Copyright 2014, John Wiley & Sons, Inc.

good biological activity. They claimed that the anti-cell adhesion property of the PET nanocone arrays resulted from the superoleophobicity, which weakened the interactions between the surfaces and secreted cell-adhesion proteins. In such a case, the inhibited formation of actin cytoskeleton led to the anti-cell adhesion property. These studies prove that the underwater superoleophobicity can greatly improve the anti-biofouling properties.

4. Bioinspired Slippery Surfaces

As same as the underwater fish scales that adopting the trapped water layers, *Nepenthes* pitcher plants use surface micro/nanotextures to lock-in an intermediary liquid and form a continuous overlying liquid film, which can effectively make the insects that step on it to slide into the digestive juices.^[59] Different to the superhydrophobic surfaces and underwater superoleophobic surfaces that prevent fouling organisms via reducing the close contact between biofouling and material surfaces, *Nepenthes* pitcher plants inspired slippery surfaces resist fouling organisms by using their stably immobilized, molecularly smooth liquid interface.^[29] For instance, by infiltrating perfluorinated lubricant into porous

polytetrafluoroethylene (PTFE) membrane, Epstein et al. presented that bioinspired slippery surfaces provided a smooth liquid-like interface to inhibit the attachment of bacteria (**Figure 4a**).^[60] Compared with the porous PTFE that accumulated biofilm within hours, the slippery surfaces suppressed 99.6% of *P. aeruginosa* at least a 7 days period under both static and physiologically realistic flow conditions, showing a 35 times greater improvement compared with the traditional PEG films. In addition, the slippery surfaces prevented 97.2% of *S. aureus* and 96% of *E. coli* from attaching onto the surfaces. They claimed that the inability of biofilm to firmly attach to the slippery surfaces and its effective removal under mild flow conditions (about 1 cm s^{-1}) were caused by their unique, non-adhesive, exceptional slippery characteristics, independent of any specific chemical or physical features of the biofouling. The slippery interfaces do not degrade over a long timescale and are stable in various harsh conditions, such as extreme pH, salinity, and UV exposure. Moreover, the slippery surfaces can be easily fabricated on arbitrarily shaped surface contours, such as catheters, pipelines, or other systems requiring prolonged resistance of biofouling.

Recently, the same research group reported a perfluorocarbon lubricant infused nanoporous tungsten oxide (TO) surface (TO-SLIP), which could be deposited on

mechanically durable steel surfaces with various shapes and showed omniphobicity.^[63] TO-SLIPS surfaces can significantly reduce the adhesion of *E. coli*, blood staining, and marine algal film, preventing bio-corrosion and fouling of steels broadly used in implants, surgical instruments, marine construction, and maritime vessels. Xiao et al. reported slippery BMA-EDMA surfaces by the infusing fluorocarbon lubricants into the microporous poly(butyl methacrylate-co-ethylene dimethacrylate) P(BMA-EDMA).^[64] The slippery BMA-EDMA surfaces exhibit remarkable inhibition of the settlement/attachment of both zoospores of the alga *Ulva linza* and cypris larvae of the barnacle *Balanus amphitrite*.

However, in another study, they found that the formation of biofilms on the slippery surfaces was highly dependent on the bacterial strains in a more relevant to physiological conditions.^[65] In low a nutrient medium, the slippery BMA-EDMA surfaces resisted the biofilm formation of different strains of opportunistic pathogen *P. aeruginosa* up to 7 days. However, in high nutrient medium, the slippery BMA-EDMA surfaces exhibited anti-biofouling property for laboratory reference clones of *P. aeruginosa* and most of the environmental *P. aeruginosa* strains. Nevertheless, for the antibiotic multi-resistant *P. aeruginosa* isolated from wastewater, dense biofilms were formed on the slippery BMA-EDMA surfaces during incubation of 7 days. Although the slippery surfaces exhibit superior anti-biofouling capability, they suggest that it is of significance to investigate both wild type and laboratory biofouling as well as different experimental conditions when evaluating the anti-biofouling capabilities of novel surface coatings.

Thrombus formation on the surfaces of extracorporeal circuits and indwelling medical devices often cause significant morbidity and mortality. Recently, bioinspired slippery surfaces have been reported to efficiently reduce thrombosis.^[61,66] Leslie et al. fabricated a tethered-liquid perfluorocarbon (TLP) based slippery surface by covalently binding a flexible molecular perfluorocarbon layer or tethered perfluorocarbon (TP) on material surfaces and subsequently coated with a mobile layer of perfluorodecalin (LP) (Figure 4b).^[61] The TLP-based slippery surfaces could effectively repel whole blood, reduce fibrin polymerization, and suppress both adhesion and activation of platelets, hence confirming their reduced thrombogenicity when contacting with flowing whole human blood in vitro. They assembled a TLP-coated AV shunt by using commercially available polyurethane cannulae, polycarbonate connectors, and PVC cardiopulmonary perfusion tubing. The TLP-coated AV shunt exhibited a remarkable decrease in occlusive thrombosis in both the arterial and venous sections when directly contacting with whole blood flowing at high, physiologically relevant rates for at least 8 h in vivo. In addition, the TLP-based slippery surfaces can be created on virtually any medical-grade material surfaces for both antithrombogenic and anti-biofouling, which show great potential in various types of extracorporeal circuits and indwelling devices.

In addition to the liquid-infused porous structure based slippery surfaces, organogel-based slippery surfaces^[62,67–71] have been also developed and utilized for anti-biofouling. Yao et al. reported lubricant-swollen fluorogels by

photocuring perfluorinated alkyl acrylate monomers (2-perfluorooctylethyl acrylate (PFOEA) or 2-perfluorohexylethyl acrylate (PFHEA)) and fluorinated macromolecular cross-linker (perfluoropolyether dimethacrylate (PFPE-DMA)) and subsequently swelling with fluorinated lubricants (FC-70).^[62] The swollen fluorogels possess wet, homogeneous, lubricated surfaces with free, dangling polymer chain ends, which can facilitate the reduction of interfacial friction and promote the slipperiness of the fluorogel surfaces (Figure 4c). Compared with the tissue culture polystyrene (TCPS) that enables good adhesion and spreading of mouse embryonic fibroblast cells, the fluorogels can significantly inhibit the cell spreading. Furthermore, the FC-70-swollen fluorogels can minimize the adhesion of blood onto their surfaces while pinning and streaking of blood on bare PFOEA-50 fluorogels, indicating the great potential as anti-biofouling surfaces.

5. Conclusion and Outlook

Here, we have suggested a superwettability-based strategy for the rational design of bioinspired anti-biofouling surfaces. Inspired by natural anti-biofouling surfaces with superwettability (such as Lotus leaf, fish scales, and *Nepenthes* pitcher plants), superhydrophobic, underwater superoleophobic, and omniphobic slippery surfaces have been successfully developed as anti-biofouling surfaces to effectively resist proteins, bacteria, cells, and marine organisms. By using the trapped air cushions or liquid layers, these surfaces can significantly prevent the close contact and adhesion between fouling organisms and material surfaces, thus leading to the excellent anti-biofouling performances. This paves the way for the further exploitation of superwettability-based anti-biofouling surfaces in biomedical and marine industrial applications.

Although significant progress has been achieved in this field, the research of bioinspired superwettability-based anti-biofouling surfaces is still at an infant stage and numerous fundamental and practical issues remain unsolved. In our opinion, future research should mainly focus on the following aspects. Firstly, more natural anti-biofouling surfaces need to be explored and discovered so that we can reveal their mechanisms and design principles of anti-biofouling and guide the design and development of novel anti-biofouling surfaces. Secondly, the development of bioinspired anti-biofouling surfaces with multiple functions, such as fouling-release, biocompatibility, high mechanical properties, and anti-abrasion, is still challenging in this field. Finally, great efforts should be made to further promote the duration and long-term performance of bioinspired anti-biofouling surfaces in practical conditions, such as high temperature, high acidity, high alkali, and high salinity to satisfy the medical, marine, and industrial applications. We envision that the novel bioinspired anti-biofouling surfaces based on superwettability strategies will bring a new era to both biomedical technology and the marine industry, and will greatly benefit human health and daily life in the near future.

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